

Classroom Attendance Prediction Using Academic Schedule and Historical Attendance Data

Domain: Machine Learning & Predictive Analytics	Dataset Scope: MCA Semesters 1, 2, 3
Dataset Size: 2,406 Lecture Records	Best Model: Random Forest (MAE: 4.17%)
Deployment: Streamlit Web UI + Pickle Bundle	Status: Validated & Ready for Submission

Chapter 1: Executive Summary & Abstract

This capstone project implements an end-to-end Machine Learning pipeline developed to predict continuous classroom attendance percentages (0.0% to 100.0%) and estimate expected physical student headcounts. By integrating academic timetables, course formats, weather conditions, exam proximity flags, and strictly leakage-free autoregressive historical attendance lags across 2,406 lecture observations in the MCA program, the system provides proactive decision support for academic administrators.

Chapter 2: Problem Statement & Objectives

Higher education institutions routinely log attendance reactively. Student turnout fluctuates significantly due to early morning scheduling, post-lunch fatigue, proximity to internal tests, assignment submission deadlines, and course delivery formats (Theory vs. Lab practicals). The objective is to construct an automated predictive system that forecasts attendance beforehand to optimize room scheduling, timetable design, and early student intervention.

Chapter 3: Dataset Architecture & Cleaning

The dataset comprises 2,406 lecture records spanning Semesters 1, 2, and 3 across Sections A & B (102-103 students). The automated cleaning pipeline (src/data_cleaning.py) enforces boundary constraints ($0 \leq \text{Present} \leq \text{Enrolled}$), mathematically recalculates the target Attendance Percentage = $(\text{Present} / \text{Enrolled}) * 100$, and verifies zero target leakage.

Chapter 4: Feature Engineering & Chronological Split

Eighty-two transformed dimensions are constructed: (1) Calendar signals (Day of Semester, Week Number, Holiday proximity, Exam week flag); (2) Timing signals (Start Hour, Time of Day, Before/After Lunch, Daily sequence); (3) Shifted Autoregressive Lags (Rolling 3-lecture average attendance per subject/cohort, Macro subject/faculty historical means). Chronological splitting (70% Train, 15% Validation, 15% Test) strictly prevents lookahead data leakage.

Chapter 5: Machine Learning Benchmarks & Selection

Five regression algorithms and seven classification algorithms were benchmarked. Random Forest Regressor achieved the highest predictive precision on the validation set (MAE: 4.170%, RMSE: 5.170%, R2: 0.3201) and maintained robust out-of-sample generalization on the untouched test holdout (MAE: 4.386%, RMSE: 5.521%, MAPE: 5.65%). Support Vector Machine (SVM) achieved 79.78% accuracy in identifying at-risk turnout.

Chapter 6: Deployment & Conclusion

The model is packaged into `attendance_model.pkl` and deployed via an interactive 7-module Streamlit web dashboard. The system enables real-time schedule entry, immediate attendance percentage and headcount estimation, timetable bottleneck diagnostics, and automated CSV register ingestion.